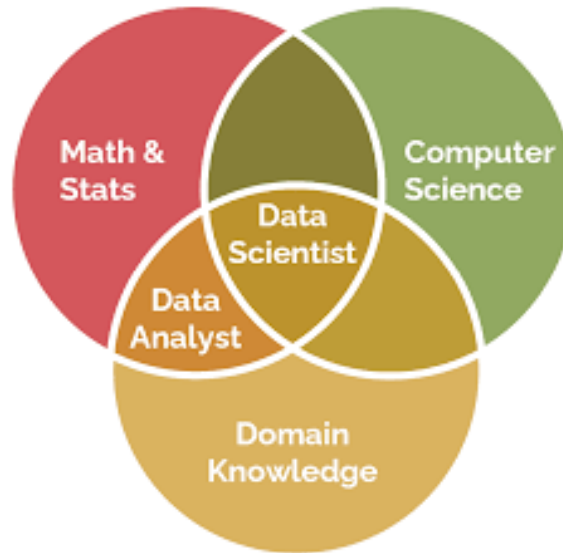


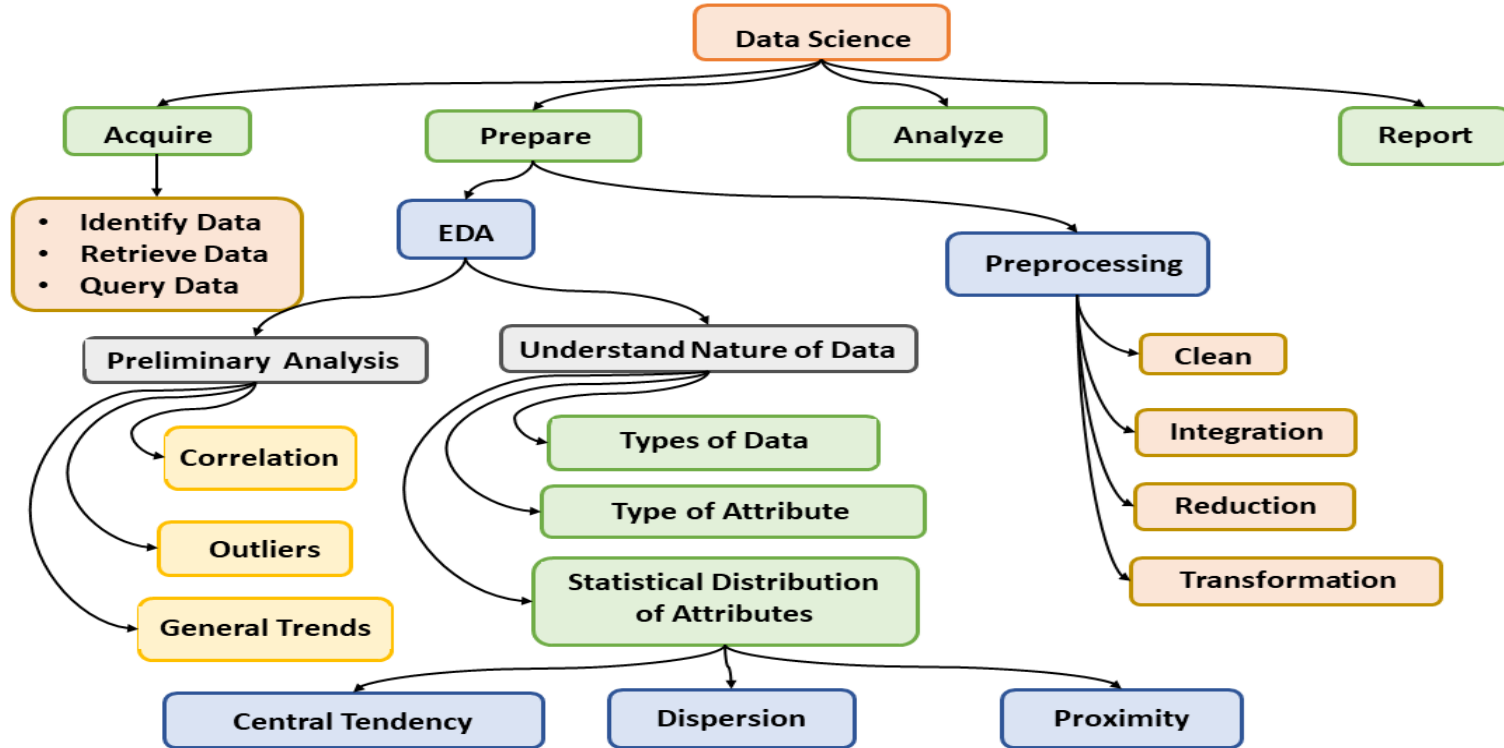
Data Science

Data Science

Data science is the field of **extracting knowledge** and insights from data through a blend of **statistics, programming, machine learning, and domain expertise**.



Data Science Processes



Types of Data

Types of Attribute

Structured Data

- Structured data is organized and follows a **predefined format**, usually stored in **databases** or **spreadsheets**.
- **Example:** A customer database with columns for name, age, email, and purchase history.
 - Ordinal data
 - Nominal data
 - Numerical data

Structured Data Examples

Ordinal:

- Movie ratings (1-5 stars)
- T-shirt sizes (S, M, L)
- School grades (A, B, C)

Nominal:

- Blood types (A, B, AB, O)
- Hair colors (blonde, brunette, redhead)
- Favorite sports (soccer, basketball, tennis)

Numerical:

- Ages (years)
- Temperatures (°C)
- Book page numbers

Bonus:

Traffic light colors (red, yellow, green)
nominal for each light, ordinal for sequence.

Unstructured Data

- Unstructured data is **not organized** and lacks a predefined format, often in the form of **text, images, audio, or video**.
- **Example:** Social media posts, customer reviews, or images from a surveillance camera.

Semi Structured Data

- Semi structured data has some organization but does not adhere to a **strict schema**, often **containing tags** or **labels**.
- **Example:** Emails, JSON files that contain data with tags or key value pairs.

Unstructured vs Structured Data



Structured Data

Often numbers or labels, stored in a structured framework of columns and rows relating to pre-set parameters.



ID CODES IN DATABASES



NUMERICAL DATA GOOGLE SHEETS



STAR RATINGS



Semi-unstructured Data

Loosely organized into categories using meta tags



EMAILS BY INBOX, SENT, DRAFT



TWEETS ORGANIZED BY HASHTAGS



FOLDERS ORGANIZED BY TOPIC



Unstructured Data

Text-heavy information that's not organized in a clearly defined framework or model.



MEDIA POSTS, EMAILS, ONLINE REVIEWS



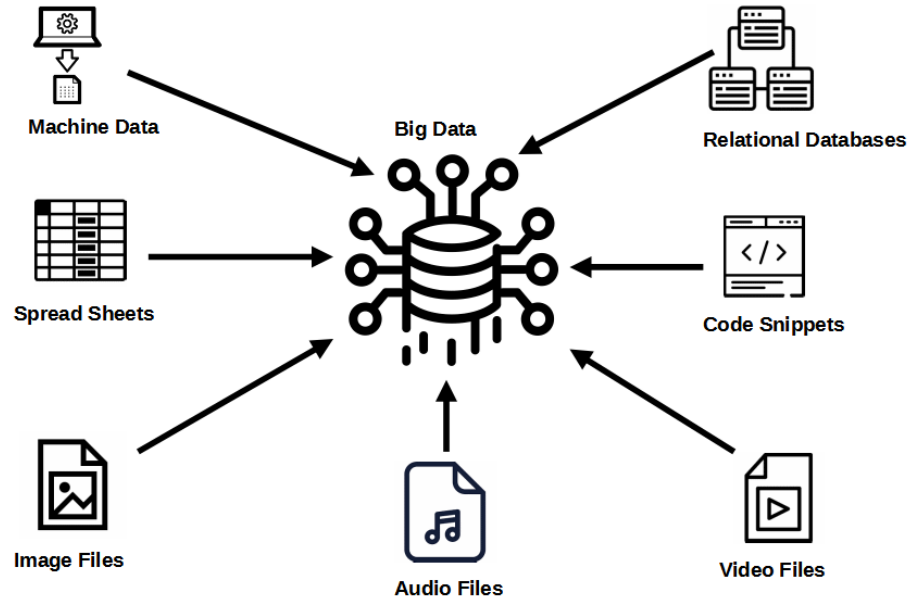
VIDEOS, IMAGES



SPEECH, SOUNDS

Data Acquisition

- **Data acquisition** refers to the **process of collecting raw data** from various sources, which could be **structured** or **unstructured**, to be used for analysis, modeling, or other purposes.
- Identify Data
- Retrieve Data
- Query Data



What is Exploratory Data Analysis (EDA)?

- The process of **exploring** and **summarizing** the main characteristics of the data to uncover **patterns**, **relationships**, and **trends**.
- It helps in formulating questions and making data driven decisions.



Importance of EDA:

- Provides an **initial understanding** of the dataset.
- Helps in identifying data **quality issues**, such as **missing values**, **outliers**, and **inconsistencies**.
- Guides the selection of appropriate statistical techniques and **models**.
- Helps in feature engineering and **variable selection**.
- Enables the discovery of **meaningful insights** and actionable conclusions.

Descriptive Statistics

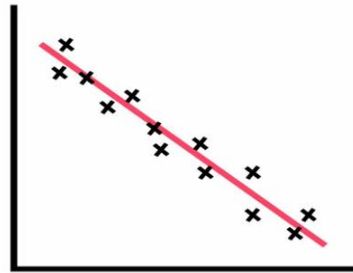
Descriptive statistics is the branch of statistics that focuses on **summarizing** and **describing** the main **features/Attributes/Variables** of a dataset.

Correlation Analysis

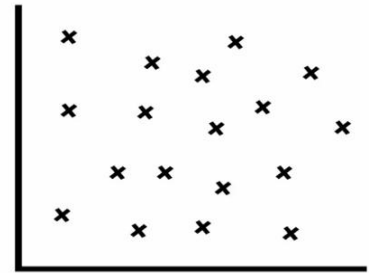
Correlation analysis is used to determine the **relationship** between **two** or **more variables** in a dataset. It helps us understand how **changes** in one variable **affect another variable**.



Positive
Correlation

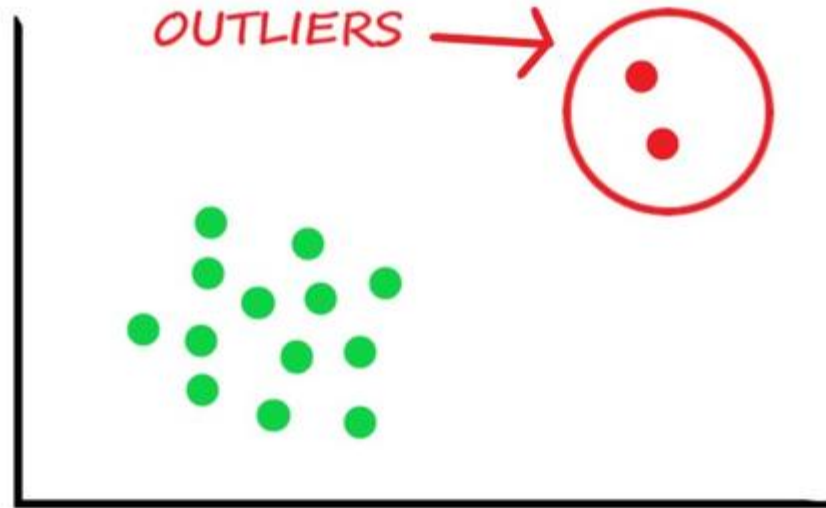


Negative
Correlation



No
Correlation

Outliers



Central Tendency

- A measure of **central tendency** is a single value that attempts to **describe** a set of data by **identifying the central position** within that set of data. As such, measures of central tendency are sometimes called measures of central location.
- The **mean**, **median** and **mode** are all valid measures of **central tendency**, but under different conditions, some measures of central tendency become more appropriate to use than others.

Mean

The mean is the **average** of a set of numbers.

Calculated by summing all the numbers in the dataset and dividing by the total count.

Formula: Mean = (Sum of all numbers) / (Total count)

Example: For the dataset [5, 7, 10, 12, 15], the mean is $(5 + 7 + 10 + 12 + 15) / 5 = 9.8$.

Sensitive to outliers

Example: For the dataset [5, 7, 10, 12, 150], the mean is $(5 + 7 + 10 + 12 + 150) / 5 = 36.8$.

Median

The median is the **middle value** in a **sorted** dataset.

If the dataset has an odd number of values, the median is the middle number.

If the dataset has an even number of values, the median is the average of the two middle numbers.

Example 1: For the dataset [3, 5, 6, 8, 9], the median is 6.

Example 2: For the dataset [2, 4, 6, 8], the median is $(4 + 6) / 2 = 5$.

Mode

The mode is the value that appears **most frequently** in a dataset.

A dataset can have one mode, more than one mode (multimodal), or no mode (no value repeats).

Example 1: For the dataset [3, 4, 4, 6, 8], the mode is 4.

Example 2: For the dataset [1, 2, 3, 4, 5], there is **no mode**.

Quartiles and Interquartile Range (IQR)

Quartiles: Before understanding the IQR, we need to grasp the concept of quartiles.

Quartiles divide a dataset into **four equal parts**, each representing a quarter (**25%**) of the data.

The first quartile (**Q₁**) is the value below which **25%** of the data falls.

The second quartile (**Q₂**) is the value below which **50%** of the data falls. It's actually **Median**.

The third quartile (**Q₃**) is the value below which **75%** of the data falls.

Finding Quartiles

To find the quartiles, first, arrange the dataset in **ascending order**.

If the dataset has an odd number of values, the median is the second quartile (Q_2).

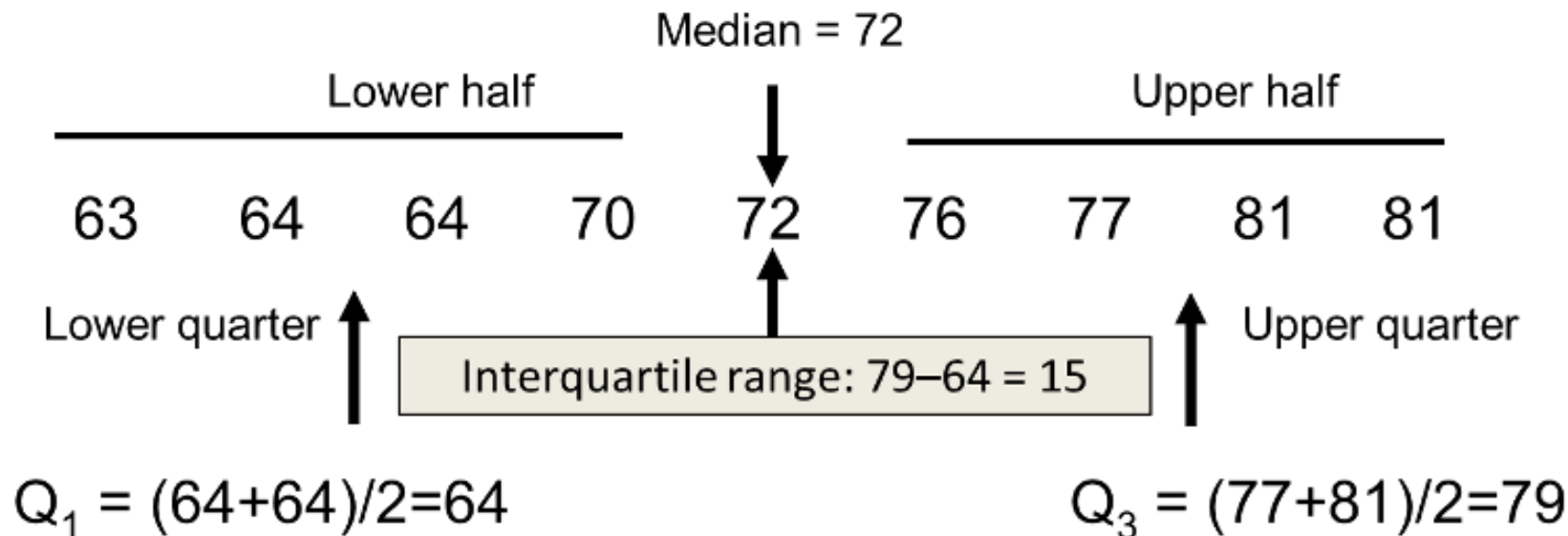
If the dataset has an even number of values, calculate the median of the lower and upper halves to find Q_2 .

Interquartile Range (IQR):

The IQR represents the difference between the first and third quartiles (Q_1 and Q_3). It indicates the spread of the middle 50% of the data.

$$\text{Formula: } \text{IQR} = Q_3 - Q_1$$

Interquartile Range (IQR):



Interpretation of the IQR

The IQR provides information about the **spread** and **variability** of the data.

A **larger IQR** indicates a **greater spread** and more variability in the dataset.

Conversely, a **smaller IQR** suggests a **less spread** and less variability.

Applications

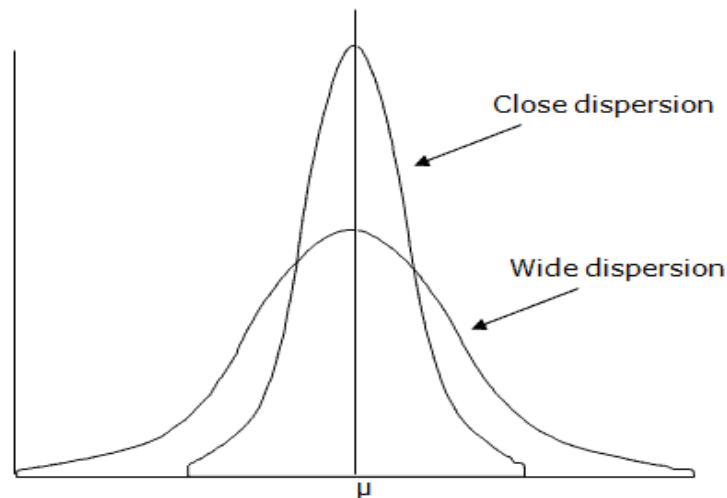
The IQR is widely used in various fields, **including economics, healthcare, and social sciences.**

It helps compare and analyze data distributions, identify **abnormal data points**, and detect **potential patterns or trends.**

Why we care about Dispersion?

Dispersion is the spread of **data points** around a **central value**, indicating variability within a dataset.

- Increased difficulty in **capturing patterns**.
- Increased risk of **overfitting**.
- Reduced predictive **accuracy**.
- Impact on **outlier handling**



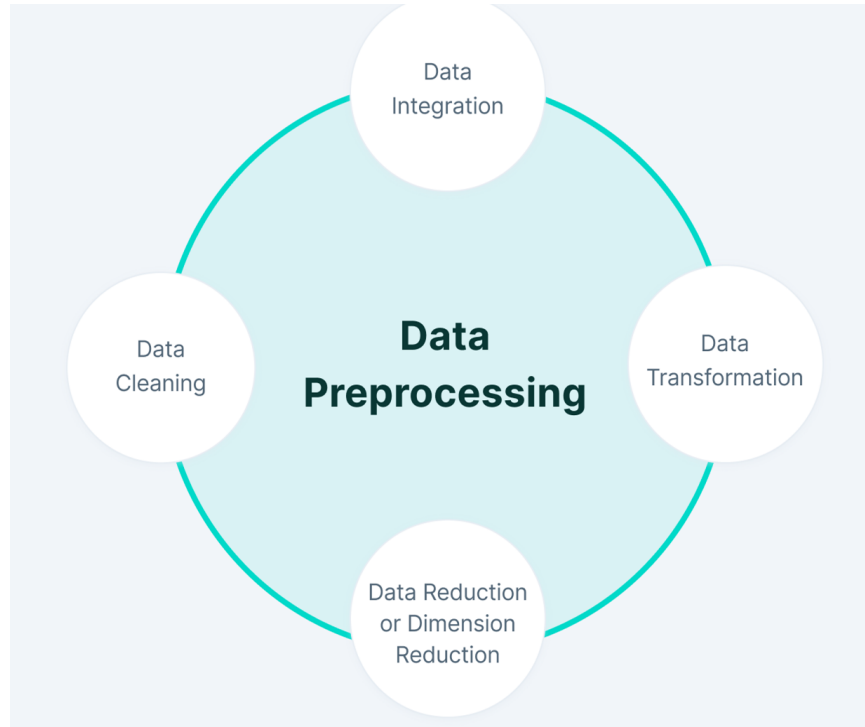
Same mean and different dispersion

Proximity:

In machine learning, "proximity" refers to the **measure of similarity** or **closeness between data points** within a dataset. It is often used in clustering algorithms and nearest neighbor methods to determine how closely related or similar two data points are to each other based on certain features or attributes.

Data Preprocessing

- **Steps:**



Data Cleaning Techniques

Handling Missing Data: We use methods like filling or removing to deal with missing values.

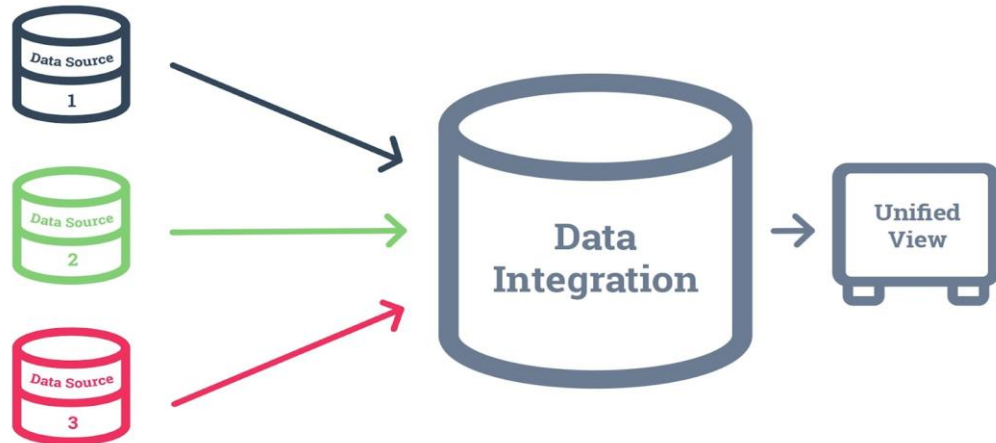
Outlier Detection: Find and address unusual data that can affect analysis or models.

Data Standardization: Make data consistent for easier analysis and comparison.

Data Validation: Check data against rules to ensure accuracy and reliability.

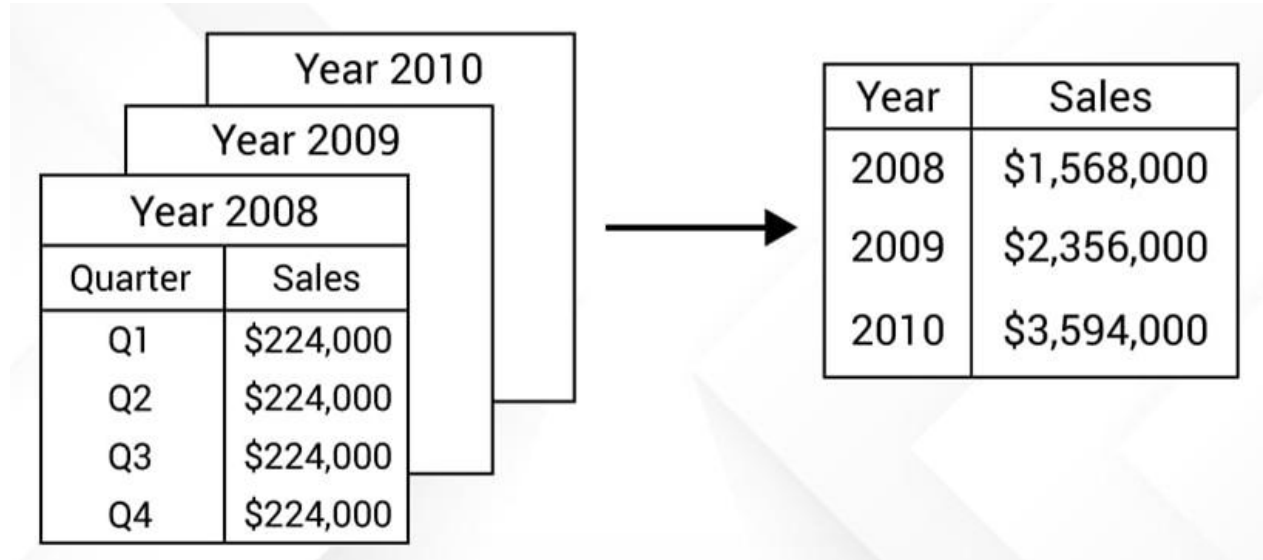
Data Integration

- Data integration is the process of **combining data** from **various sources** into a single dataset. The **goal** is to provide users with **consistent access** to data across different subjects and structure types.



Data Reduction

- Data reduction is the **process of transforming digital information into a simplified, ordered, and corrected form**. It can be used on numerical or alphabetical information.



PCA

Principal component analysis (**PCA**) is a **statistical method that reduces a data table to its main features**. It's a dimensionality reduction technique that's often used to reduce the size of large data sets. PCA works by **linearly transforming data** onto a new coordinate system.

Dimensionality Reduction Technique:

- Eigenvalue Decomposition (**Eigendecomposition**):
- Singular Value Decomposition (**SVD**):
- **t-SNE (t-Distributed Stochastic Neighbor Embedding)**

Three Main Types Of Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

What is it?

The model learns from **labeled data**, meaning each input is paired with the correct output (label).

How does it work?

The algorithm is trained on this labeled data, and it learns to predict the output for new, unseen data based on patterns it finds.

Example: Predicting house prices based on features like size, location, and condition.

Supervised Learning

Problems it solves:

Classification Problems: Where you want to categorize data into classes or groups.

Example: Email spam detection (spam or not spam).

Regression Problems: Where you predict a continuous output based on input data.

Example: Predicting house prices based on features like size, location, etc.

Unsupervised Learning

What is it?

The model works with **unlabeled data**, meaning no correct answers (labels) are given. The model must find patterns or structures on its own.

How does it work?

The model looks for hidden patterns or groupings in the data.

Example: Customer segmentation, where the algorithm groups customers based on their buying behavior without knowing the specific categories in advance.

Reinforcement Learning

What is it?

This learning type is based on agents learning through **trial and error** by receiving rewards or penalties.

How does it work?

The agent takes actions in an environment, and based on the actions, it receives feedback (positive or negative). Over time, it learns the best strategies to maximize rewards.

Example: Training a robot to navigate through a maze by rewarding it for reaching the goal and penalizing it for hitting walls.